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Artificial Intelligence (AI)

A risk-assessment framework for responsible AI use in research publishing

Springer Nature supports the responsible and transparent use of artificial intelligence where it strengthens research quality, upholds editorial independence, and preserves the integrity of the scientific and scholarly record.

These policies set out a risk-assessment framework governing how AI may be used across the research and publishing lifecycle, applied consistently to authors, peer reviewers, and editors, while recognising their distinct roles and responsibilities.

AI is treated as a supporting technology. Scholarly judgement, accountability, and responsibility always remain human.

Core expectations

Across all editorial roles and activities:

- Human accountability is non-transferable
Accountability for scholarly content, evaluation, and editorial decisions cannot be delegated to AI systems.
- AI may support, but must not replace, scholarly judgement
AI can assist clarity, efficiency, and exploration, but must not determine conclusions, evaluations, or decisions.

- Transparency creates trust and confidence
Transparent declaration of AI builds trust and confidence, and removes ambiguity about how AI might have been used.
- Confidentiality and data protection are mandatory
Manuscripts, peer review reports, and sensitive data must not be shared with unsecured or public AI systems.

Risk-assessment framework for AI use and declaration

In support of the policy, the following framework provides guidance on how to use AI safely. Use of AI is governed according to risk level, not by tool type. The same framework applies across all roles: authors, reviewers and editors.

Risk assessment level	 Green-Permitted: Assistive AI use (low risk)
Summary	AI use that supports expression, organisation, or efficiency without influencing scientific, scholarly or evaluative judgement.
Expectations	AI use is likely to be reversible and verifiable AI use does not introduce new intellectual content Accountability for scholarly or evaluative judgement remains clearly human
Examples	Using an LLM to polish or refine the language Suggesting structure or formatting of manuscript sections

	Translation, structuring or clarifying reviewer comments Comparing methodological options Stress-testing research questions Data cleaning and deduplication
Compliance and disclosure	Permitted. Disclosure enhances trust and transparency and clearly demonstrates human accountability.

Risk assessment level	 Amber - Exercise caution: Evaluative or interpretive AI use (requires caution and care)
Summary	AI use that may influence interpretation, framing, emphasis, or evaluative judgement, but remains under human control.
Expectations	AI contributes to reasoning or critique AI does introduce new intellectual content and requires verification and oversight Human judgement and accountability must be demonstrable
Examples	Suggesting analytical, experimental or methodological approaches Drafting explanatory summaries Comparing results to existing literature Extensive copy editing or writing support

	Pattern identification in exploratory data analysis Explaining outputs from statistical models in plain language Recommending statistical tests or modelling approaches
Compliance and disclosure	Permitted with human oversight, verification, and transparency through disclosure. If AI materially influences evaluative judgement, accountability must remain clearly human-led and defensible.

Risk assessment level	 Red - Not permitted: AI replacing scholarly judgement or lacking transparency
Summary	AI use that is opaque or replaces accountable human contribution, generates unverifiable outputs, or compromises confidentiality or integrity.
Expectations	AI use is opaque or creates non-verifiable outputs Author, reviewer or editorial responsibility has been delegated to AI AI use has breached confidentiality or consent
Examples	Generating hypotheses, analyses or conclusions and presenting them as human-derived Fabricating data, citations or results Using an LLM to generate core research reasoning without disclosure

	Assigning authorship or accountability to AI systems or tools Delegating peer review to an LLM Creating photorealistic images (deepfakes)
Compliance and disclosure	Not permitted.

Role-specific responsibilities

The same risk framework applies to all roles, with different responsibilities.

Authors

- Remain fully accountable for originality, accuracy, and integrity
- Disclose AI use for trust and transparency

Peer reviewers

- Remain responsible for independent, expert evaluation
- May use AI to support peer review but must not upload manuscript content to unsecured or public AI tools
- Disclose AI use for trust and transparency
- Must not delegate critique or judgement to AI systems

Editors

- Remain accountable for editorial decisions and oversight
- Apply proportional, risk-based judgement when assessing AI use
- Intervene where accountability or transparency is unclear

- Escalate to their publishing contact where integrity or policy boundaries are breached

Editorial use

Nature Portfolio journals occasionally use artificial intelligence tools developed or approved by Springer Nature to support different tasks in the generation of accessory content or to support the editorial process. Accessory content can include (but is not limited to): key points, glossary terms, editorial summaries, plain language summaries and social media posts. Accessory content is always edited and fact-checked by the author and/or editor to meet Nature Portfolio publication standards. In the editorial process, artificial intelligence is only used to summarise, retrieve or highlight information. All decisions are made by editors. Any substantive use of artificial intelligence beyond the above will be declared on an individual article basis.

For additional information on the use of artificial intelligence at Springer Nature, please visit the [Springer Nature AI Hub](#).

Governance and review

Existing research integrity, ethics, and editorial governance processes apply to all AI-related concerns. These policies will be reviewed periodically to ensure alignment with evolving technologies, regulations, and community expectations.

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